

SPECIAL PACKAGING INSTRUCTION					Form Approved OMB No. 0704-0188		XTH JS/JO
The public reporting burden for this collection of information is estimated to average 30 days per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing the burden, to the Department of Defense, Executive Services and Communications Directorate (0704-0188). Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number. PLEASE DO NOT RETURN YOUR COMPLETED FORM TO THE ABOVE ORGANIZATION.							
1. PART OR DRAWING NO. (CAGE CODE) NOMENCLATURE TIRE & WHEEL ASSEMBLY				2. CAGE 19207		3. SPI NO. AK00000001	
4. NATIONAL STOCK NO.				5. DATE 11/13/2019		6. REVISION D	
7. QUP 001		8. ICQ NONE		9. UNIT PACK WT (LB.)		10. UNIT PACK CU (CU. FT.)	
12. MILITARY PRESERVATION MIL-STD-2073-1, METHOD 10		18. STEPS		19. REQD		20. DESCRIPTION	
13. CLEANING MIL-STD-2073-1		1				GENERAL REQUIREMENTS: SEE NOTE A	
14. DRYING MIL-STD-2073-1		2				PRESERVATION: SEE NOTE D	
15. PACKING		3				REFERENCE: INTERNATIONAL AIR TRANSPORT	
a. LEVEL A MIL-STD-2073-1AND NOTE C						ASSOCIATION DANGEROUS GOODS REGULATIONS -	
b. LEVEL B MIL-STD-2073-1AND NOTE B						IATA – TIRE AND WHEEL ASSEMBLIES MAY BE	
16. MARKING MIL-STD-2073-1AND NOTE E						FORBIDDEN FOR PASSENGER AND CARGO AIR	
						MOVEMENT. SEE INSTRUCTIONS IN NOTE A-10.	
17. NOTES/DRAWING							
A: General Packaging Requirement – Note: This special packaging instruction does not apply to DRMO shipments. The marking and palletizing of T&WA's for disposal should be of absolute minimum to provide safety of personnel and should comply with local laws and DRMO REGULATIONS. 1. Army tire and wheel assemblies (T&WA) require special marking and handling. These instructions apply to all T&WA; new, repaired, as well as unserviceable/reparable. If any deviation is required from these instructions a waiver must be obtained from usarmy.detroit.tacom.mbx.ilsc-packaging@mail.mil. 2. Assemblies may be shipped horizontal (on the sidewalls) with these provisions: a. It is the shipper's responsibility to ensure assemblies are not crushed, that the bead is not broken, and that the valve assembly is not damaged due to stacking. b. The shipper is responsible to ensure shipping surfaces are free from anything that could damage the tire sidewalls, e.g. nails, wood-slivers. c. No load overhang is permitted on the pallet. 3. T&WA are easily damaged by anything sharp or protruding. Prior to loading any conveyance, all surfaces will be inspected to ensure nothing will puncture or damage the tires. When surfaces are in such condition that they may cause damage to the tires they must be lined for protection. Do not ship tire assemblies next to, on top of, or under any material that could cause damage to the assembly.							

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4. Pallets designed for stacking (e.g.; wood framed or steel framed) that support the weight of the pallet above may be stacked (see top of page 7). Otherwise, pallets may not be stacked, see page 6.

5. The tire bead, valve, and the wheel are areas that are easily damaged. Tire assemblies should never be lifted with the forks of a forklift unless the forklift has special forks designed to lift the assembly. Split Rims are especially dangerous and require careful handling.

6. Special Handling Equipment should be considered when large tire assemblies and large quantities of assemblies will be handled, stored, and issued from a facility. Shipper is responsible for notifying the customer of any special handling equipment required.

a. Before inflating/deflating the tires, stand a minimum of ten (10) feet away from the wheel on the side facing the tire tread area. The danger zone (trajectory path) extends anywhere outside the front or back of the rim of areas facing the tire sidewalls. Ensure that no one is in the danger zone (trajectory path) during the tire inflation/deflation process.

b. Inflate bolt-together and single-piece tire and wheel assemblies to normal operating pressure in an OSHA approved inflation safety cage. After inflation, carefully check the final seat of the bead and rim components before removing the assembly from the tire cage. Disconnect the inflation hose from the tire and proceed with remaining tires using the same precautions.

c. When other multi-piece wheel assemblies, e.g. split-ring, are built, they shall be inflated between 20 to 40 psi in an OSHA approved inflation safety cage so that the bead is seated. If the bead is not seated at 40 psi, the assembly needs to be completely deflated and checked for anomalies. Deflate by removing the valve core while the tire is still in the inflation safety cage. Inflate these T&WA to a maximum of 40 psi for shipment and storage.

7. All wood containers shall be lined with fiberboard, ASTM D4727, Corrugated, Domestic, Single Wall, 175 lb. bursting strength minimum. All wood containers will have skids for ease of mechanical handling.

8. Tire & Wheel Assemblies that are palletized/packed in quantities of more than one shall be separated by polypropylene foam, A-A-59135, Class 2, or corrugated fiberboard, ASTM D4727, single wall, domestic grade. This is to minimize any chafing and damage to the tire or metal to metal contact of the wheels. Damage can occur when sidewalls contact any shipping surface that may cut, puncture, or gouge the tire or damage the wheel.

9. When wood containers are used, **do not use nails for closing the container.**

10. The bolt-together and single-piece tire and wheel assemblies shall be inflated to the normal operating pressure in an OSHA approved inflation safety cage for shipment, storage and handling. The split-ring assemblies shall be inflated until the bead is seated, between 20 and 40 psi in an OSHA approved inflation safety cage for shipment, storage and handling. If the bead is not seated at 40 psi, the assembly needs to be completely deflated and checked for anomalies. Tire & Wheel assemblies are FORBIDDEN on passenger and cargo aircraft under the following conditions (REFERENCE IATA DANGEROUS GOODS REGULATIONS):

a. Tire assemblies are inflated above maximum operating pressure.

b. Unserviceable tire assemblies are pressurized.

c. Damaged tire assemblies are pressurized.

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11. The following are some generic guidelines to safely inflate T&WA. For specific instructions on assembly/disassembly of wheels and tires, refer to the latest version of applicable system Technical Manual, or applicable drawing, e.g. TM 9-2320-280-20-2 is for HMMWV vehicles.

a. Make sure to follow all proper safety procedures when assembling wheels/tires. Never re-inflate a tire that has been run flat or seriously under-inflated without removing and repairing the tire, tube, and/or both.

b. Prior to removing tires for service, and disassembly of rim and wheel components, be absolutely certain that tire is completely deflated by removing the valve stem.

c. All Wheel Assemblies: Inflate tire/wheel assembly only in an approved OSHA safety cage using a clip-on chuck and pneumatic inflator gauge, attached to a 10 foot or longer air hose. Tire Safety Cage, NSN 4910-01-373-0267, size 40-3/4 inch long, 25-inch wide, 56-inch high will fit most tactical vehicle tires. Safety cage, NSN 4910-00-025-0623, size 78-3/4 inch long, 35-1/2 inch wide, 86-1/4 inch high, is available for larger tires.

d. Utilize OSHA 1910 and AR 750-1 compliant Universal Ground-Vehicle High-Flow Safety Tire Inflator Kit 17-6F-506, 4910-01-668-5017, with 12' hose and clip-on-chuck. This kit allows for remote inflation/deflation at a safe distance to mitigate explosion injury. Completely read the guidebook contained within the kit for safety information and operational instructions.

Note: Some vehicle systems may already have this kit in their Additional Authorized List (AAL).

e. Before inflating/deflating the tires, stand a minimum of ten (10) feet away from the wheel on the side facing the tire tread area. The danger zone (trajectory path) extends anywhere outside the front or back of the rim of areas facing the tire sidewalls. Ensure that no one is in the danger zone (trajectory path) during the tire inflation/deflation process.

f. Inflate bolt-together and single-piece tire and wheel assemblies to normal operating pressure in an OSHA approved inflation safety cage. After inflation, carefully check the final seat of the bead and rim components before removing the assembly from the tire cage. Disconnect the inflation hose from the tire and proceed with remaining tires using the same precautions.

g. When other multi-piece wheel assemblies, e.g. split-ring, are built, they shall be inflated between 20 to 40 psi in an OSHA approved inflation safety cage so that the bead is seated. If the bead is not seated at 40 psi, the assembly needs to be completely deflated and checked for anomalies. Deflate by removing the valve core while the tire is still in the inflation safety cage. Inflate these T&WA to a maximum of 40 psi for shipment and storage.

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B. Level B Packing: Tire and Wheel Assemblies being shipped on their tread or shipped in a horizontal position (on-the-sidewall) shall not hang over the edge of a pallet. A standard 40 x 48 pallet shall not be used for Tire and Wheel Assemblies exceeding 39 inches in diameter, oriented in a horizontal position or those over 47 inches in diameter being oriented on their tread. 1. All palletized loads will follow the requirement in MIL-STD-147. T&WA shipped by air or OCONUS, not containerized, will have a wood/plywood cap to ensure protection to the valve assembly, the bead and rim, and for safety and compliance with special provisions of IATA DGR. 2. Framed and capped palletized loads of T&WAs that are shipped in the horizontal position (on-the-sidewalls) on any pallet shall not exceed 55 inches in height, including the pallet height. 3. Fabricate pallets large enough to accommodate the size of the tire and wheel assemblies. To the maximum extent possible use standard size pallets. 4. When shipping tire and wheel assemblies palletized vertically (on the tread) place fiberboard separators, see note A8, between each assembly. When shrink wrapping or stretch wrapping is used, placing sheets of polyethylene between each assembly is an acceptable alternative. This is to eliminate any chafing and damage that may be caused by metal to metal contact of the wheels. Filament reinforced strapping tape may be used around the circumference of the load prior to shrink/stretch wrapping, see top of page 7. 5. Pallet loads will not be stacked one on top of another unless the load is framed and capped. When shipping tire and wheel assemblies palletized horizontally (on the sidewall), place fiberboard separators, see note A8, between each T&WA. Use of a top and bottom sheet of fiberboard is mandatory to provide minimum protection when shipping T & WAs horizontally. 6. Pallet shall comply with ASME MH1. As a general rule pallets shall be Stringer or block, limited use, partial or full four-way entry, wing type, double deck design. If flush pallets are used, the stretch or shrink wrapped pallet load of tire/wheel assemblies must be further secured by banding, to include fibreboard edge protectors used in areas to keep banding from damaging the tires , as shown as page 6.. Types of wood pallets intended for military use are provided in ASME MH1, Part 9. These pallets are commonly used by DoD activities for the handling, storage, and transportation of materials. 7. Polyethylene (PE) shrink wrap for use in pallet load securement is to conform to type IV, class 3, grade A, finish 1, of A-A-3174. The film will be a bag encapsulated over the entire load and will extend at least evenly with the underside of the bottom deck board of the pallet. PVC films shall not be used.		

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8. Stretch wrap is appropriate for palletized loads of tire and wheel assemblies shipped in the continental United States (CONUS) or when shipped containerized. Wing type pallets shall be used in order to properly anchor the load to the pallet. If a flush type pallet is used, further securement of the load to the pallet by banding is required, to include fibreboard protectors (see note A.8.). Stretch wrap should be clear, extruded PE, 0.9 mil minimum thickness, or Ethylene Vinyl Acetate (EVA), 0.8 mil minimum thickness. Multiple wraps of PE need to add up to a minimum of 2.7 mils thickness, prior to wrapping, on loads up to 1,000 pounds, 4.5 mils thickness on loads up to 2,000 pounds, and 5.6 mils thickness on loads up to 3,000 pounds. When using EVA film, a minimum thickness of 1.6 mils is required for loads up to 1,000 pounds, 2.4 mils thickness for loads up to 2,000 pounds, and 3.2 mils thickness for loads up to 3,000 pounds. Film is to be wrapped around the load in multiple wraps, either by the spiral, full wrap method or by the push-through method. Wrap the load from the top of the bottom deck board to not less than 2 inches above the height of the load. To provide additional protection, a sheet of fiberboard, see Note A8, the same size as the perimeter of the load, should be placed on top and bottom of the load prior to wrapping. When handheld, stretch-wrap equipment is used, ensure that a uniform tension exists throughout the load.

C. Level A Packing: Shippers and carriers are responsible to ensure tire and wheel assemblies are properly prepared and protected for shipment.

1. When shipping and storing T&WA's packed in a container vertically (on the tread) place fiberboard separators, see Note A8, between each T&WA. This is to eliminate any chafing and damage that may be caused by metal to metal contact of the wheels.

2. If shipping tire and wheel assemblies horizontally, only pack one T&WA to a container. Damage can occur when the sidewalls contact any shipping surface that may cut, puncture, or gouge the tire or damage the wheel. Line all shipping containers with fiberboard, ASTM D4727, see Note A11 above. The tire sidewalls must be protected from damage by the use of a protective sheet of fiberboard between the tire and wheel assembly and the container to eliminate damage.

3. Authorized shipping containers include ASTMD6880, ASTM D6251, ASTM D6256, ASTM D6039, ASTM D6254, or ASTM D6573. The sectional pallet box with one each T&WA per section is an acceptable pack for shipping and storing, see MIL-STD-147. It is acceptable to use skids and rubbing strips in lieu of the pallet base for the sectional pallet box.

D. When specified, apply A-A-52408 to external tire (rubber) surfaces only. This preservative is intended to provide long term storage protection from ozone and UV deterioration to the rubber.

E. Special Marking:

1. The following marking shall be applied to each Level B pallet load, as appropriate:

a. "DO NOT USE KNIVES, RAZORS, OR ANYTHING SHARP TO UNPACK OR PACK TIRE ASSEMBLIES."

b. DO NOT SHIP TIRES NEXT TO, ON TOP OF, OR UNDER ANY MATERIAL THAT COULD CAUSE DAMAGE TO THE TIRE ASSEMBLY.

c. PALLETS MAY NOT BE STACKED UNLESS A FRAME STRUCTURE IS USED TO SUPPORT THE LOAD.

d. DEFLATE ALL UNSERVICEABLES PRIOR TO TURN-IN.

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2. Marking of tires can only be accomplished through the use of adhesive backed labels or envelopes that are applied to the tread. No adhesive that will degrade the tire should be used to attach labels.

3. Split rim assemblies can be a transportation hazard. All split rim assemblies, see table II, will contain this additional markings: "DANGER: SPLIT RIM TIRE AND WHEEL ASSEMBLY IS UNDER PRESSURE – HANDLE WITH CARE-DO NOT INFLATE/DEFLATE EXCEPT IN AN APPROVED MANNER"

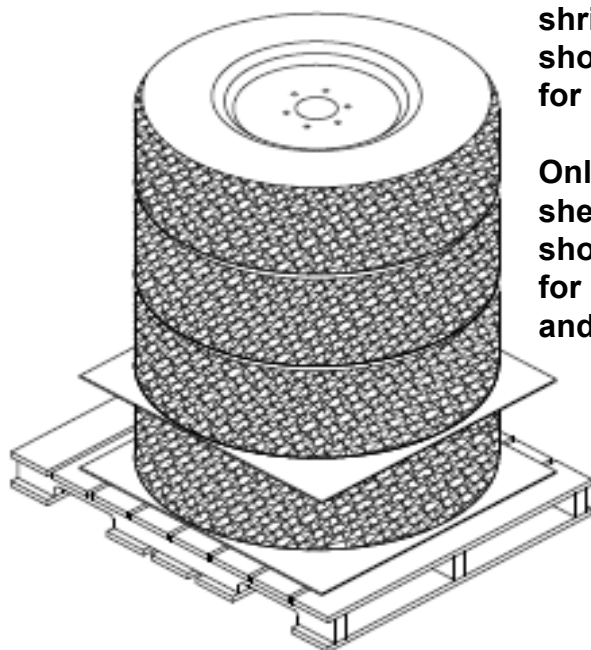
4. **SHELF LIFE – TIRE AND WHEEL ASSEMBLIES HAVE AN EXTENDIBLE SHELF LIFE. THE SPECIFIC SHELF LIFE MARKING MUST BE APPLIED IN ACCORDANCE WITH MIL-STD-129. TIRE AND WHEEL ASSEMBLY MUST HAVE THEIR TIRE REPLACED WITH A NEW TIRE AS PART OF EXTENDING SHELF LIFE.**

Note:

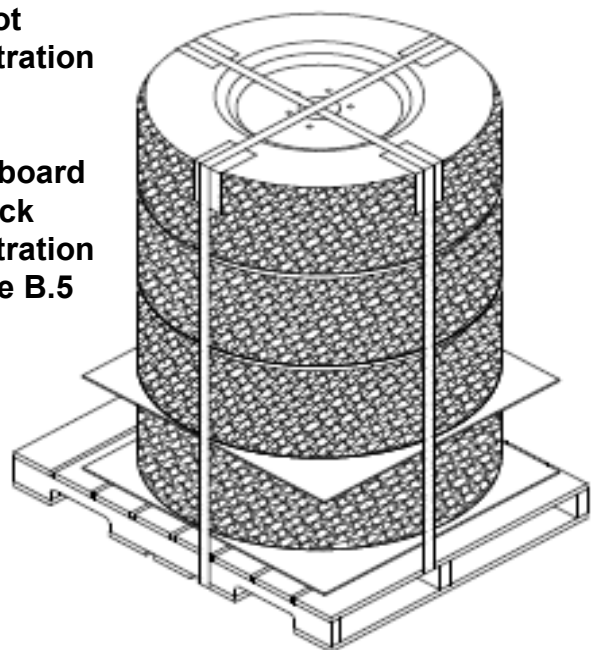
ILLUSTRATION BELOW DOES NOT DEPICT THE FRAMED AND CAPPED LOAD OPTION, REQUIRED IN ORDER TO STACK PALLETS. ONLY FRAMED AND CAPPED LOADS MAY BE STACKED. (SEE B.5)

Stretch wrap or shrink wrap not shown in illustration for clarity.

Only two fiberboard sheets per stack shown in illustration for clarity. (See B.5 and A.8)



WINGED PALLET



FLUSH PALLET

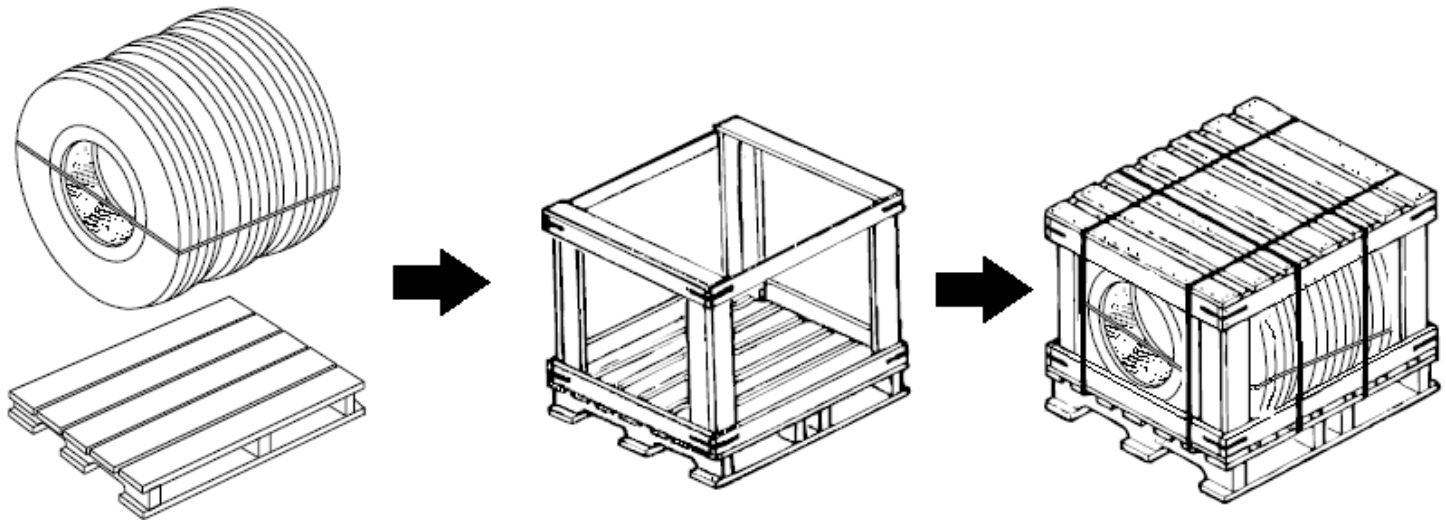
Acceptable Level B Packing

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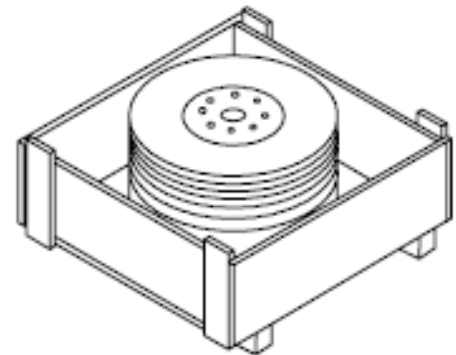
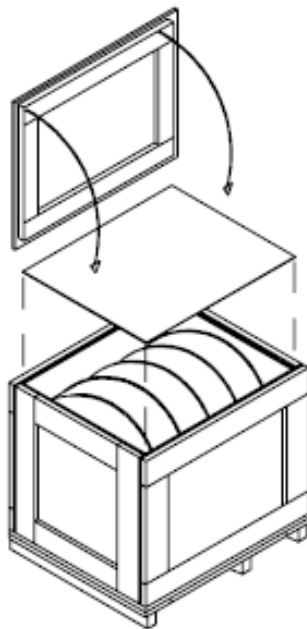
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LEVEL B PACKS



LEVEL A PACKS

